MALAWI UNIVERSITY OF BUSINESS AND APPLIED SCIENCES

SCHOOL OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF MATHEMATICAL SCIENCES EDUCATION

BACHELOR OF SCIENCES IN MATHEMATICAL SCIENCES

YEAR THREE

DATA STRUCTURES AND ALGORITHMS

CIS-DSA-321

ASSIGNMENT 1

ASSIGNMENT TITLE : MALAWI ROAD NETWORK

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QUESTION ONE

Scenario: A road network joining 20 districts of Malawi, listed from the far north (A, **Chitipa**) to the far south (T, Blantyre). Each district is a vertex, each road between two districts is an edge, and the weight of an edge is the distance in km.

Type: Weighted, **undirected** graph (a road can be travelled in both directions, so the edges have no arrowheads).

Vertices (V = 20) and edges (E = 32). The letters below are the same letters used in the hand-drawn graph.

Letter	District	Region	Letter	District	Region
A	Chitipa	North	K	Salima	Central
B	Karonga	North	L	Lilongwe	Central
C	Rumphi	North	M	Dedza	Central
D	Nkhata Bay	North	N	Mangochi	South
E	Mzimba	North	O	Machinga	South
F	Nkhotakota	Central	P	Ntcheu	Central
G	Kasungu	Central	Q	Zomba	South
H	Ntchisi	Central	R	Neno	South
I	Mchinji	Central	S	Chiradzulu	South
J	Dowa	Central	T	Blantyre	South

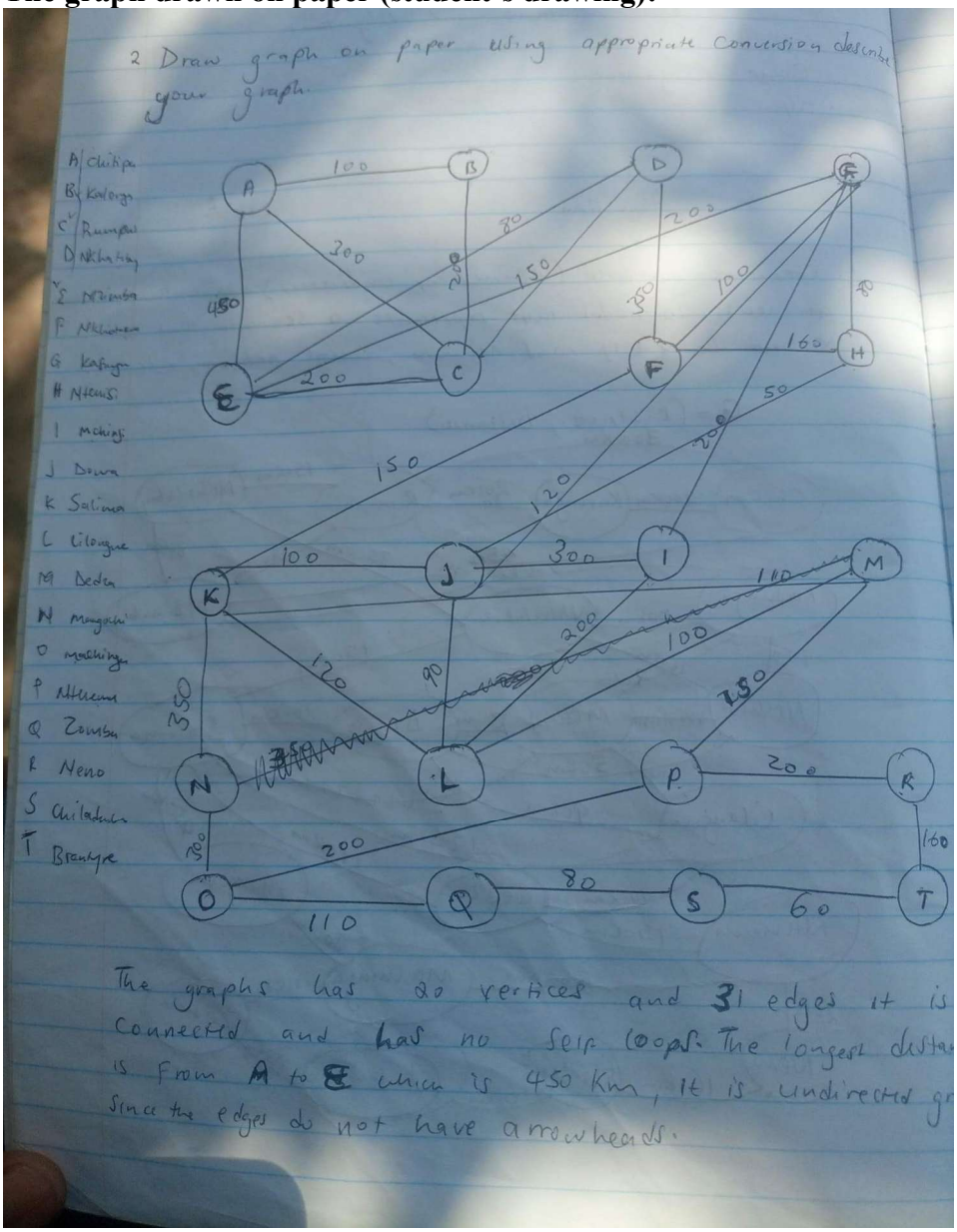
CONNECTION BETWEEN DISTRICTS

Road	Between	km	Road	Between	km
A – B	Chitipa – Karonga	100	I – J	Mchinji – Dowa	300
A – C	Chitipa – Rumphi	300	I – L	Mchinji – Lilongwe	200
A – E	Chitipa – Mzimba	450	J – K	Dowa – Salima	100
B – C	Karonga – Rumphi	200	J – L	Dowa – Lilongwe	90
C – D	Rumphi – Nkhata Bay	150	K – L	Salima – Lilongwe	120
C – E	Rumphi – Mzimba	200	K – M	Salima – Dedza	110
D – E	Nkhata Bay – Mzimba	80	K – N	Salima – Mangochi	350
D – F	Nkhata Bay – Nkhotakota	350	L – M	Lilongwe – Dedza	100
E – G	Mzimba – Kasungu	200	M – P	Dedza – Ntcheu	150
F – G	Nkhotakota – Kasungu	100	N – O	Mangochi – Machinga	300
F – H	Nkhotakota – Ntchisi	160	O – P	Machinga – Ntcheu	200
F – K	Nkhotakota – Salima	150	O – Q	Machinga – Zomba	110

Road	Between	km	Road	Between	km
G – H	Kasungu – Ntchisi	80	P – R	Ntcheu – Neno	200
G – I	Kasungu – Mchinji	200	Q – S	Zomba – Chiradzulu	80
G – J	Kasungu – Dowa	120	R – T	Neno – Blantyre	160
H – J	Ntchisi – Dowa	50	S – T	Chiradzulu – Blantyre	60

QUESTION TWO

The graph drawn on paper (student's drawing):



Description of the graph:

- Each district is a circle with its letter inside; the key on the left of the page gives the district name.
- Each road is a straight line joining two circles, and the number on the line is the distance in km.
- The graph is **undirected** because no arrows are drawn
- It has **20 vertices and 32 edges**, is **connected** (every district can be reached from every other district) and has **no self-loops** and no repeated roads.
- It is undirected. The sum of all degrees is 64, which is twice the number of edges, as it should be.
- The **longest road** is A – E (Chitipa – Mzimba) at **450 km**. The **shortest road** is H – J (Ntchisi – Dowa) at **50 km**.
- The districts with the most roads (degree 5) are G (Kasungu), J (Dowa) and K (Salima). The districts with the fewest roads (degree 2) are B, N, Q, R, S and T. No vertex has degree 1.
- It contains many cycles, for example F – G – J – K – F, J – K – L – J and O – Q – S – T – R – P – O.

QUESTION THREE

Dijkstra finds the shortest distance from one source to every other vertex when all weights are non-negative. if dist[u] + w < dist[v], then dist[v] is updated.

set dist[source] = 0 and all others to infinity; put the source in a min-priority queue; while the queue is not empty, pop the smallest vertex u; for each neighbour v, relax the edge (u, v) and push v if improved.

source = A (Chitipa)

Node	DIST	PRED
A	0	—
B	100	A
C	300	A
D	450	C
E	450	A
F	750	G
G	650	E
H	730	G
I	850	G
J	770	G
K	870	J
L	860	J
M	960	L
N	1220	K
O	1310	P
P	1110	M

Q	1420	O
R	1310	P
S	1500	Q
T	1470	R

Iteration 1: Node A • **B:** 0 + 100 = 100 • **C:** 0 + 300 = 300 • **E:** 0 + 450 = 450	**Iteration 2: Node B** • **A:** 100 + 100 = 200 • **C:** 100 + 200 = 300
Iteration 3: Node C • **A:** 300 + 300 = 600 • **B:** 300 + 200 = 500 • **D:** 300 + 150 = 450 • **E:** 300 + 200 = 500	**Iteration 4: Node E** • **A:** 450 + 450 = 900 • **C:** 450 + 200 = 650 • **D:** 450 + 80 = 530 • **G:** 450 + 200 = 650
Iteration 5: Node D (dist 450) • **C:** 450 + 150 = 600 • **E:** 450 + 80 = 530 • **F:** 450 + 350 = 800	**Iteration 6: Node G** • **E:** 650 + 200 = 850 • **F:** 650 + 100 = 750 • **H:** 650 + 80 = 730 • **I:** 650 + 200 = 850 • **J:** 650 + 120 = 770
Iteration 7: Node H • **F:** 730 + 160 = 890 • **G:** 730 + 80 = 810 • **J:** 730 + 50 = 780	**Iteration 8: Node F** • **D:** 750 + 350 = 1100 • **G:** 750 + 100 = 850 • **H:** 750 + 160 = 910 • **K:** 750 + 150 = 900
Iteration 9: Node J • **G:** 770 + 120 = 890 • **H:** 770 + 50 = 820 • **I:** 770 + 300 = 1070 • **K:** 770 + 100 = 870 • **L:** 770 + 90 = 860	**Iteration 10: Node I** • **G:** 850 + 200 = 1050 • **J:** 850 + 300 = 1150 • **L:** 850 + 200 = 1050
Iteration 11: Node L • **I:** 860 + 200 = 1060 • **J:** 860 + 90 = 950 • **K:** 860 + 120 = 980 • **M:** 860 + 100 = 960	**Iteration 12: Node K** • **F:** 870 + 150 = 1020 • **J:** 870 + 100 = 970 • **L:** 870 + 120 = 990 • **M:** 870 + 110 = 980 • **N:** 870 + 350 = 1220
Iteration 13: Node M • **K:** 960 + 110 = 1070 • **L:** 960 + 100 = 1060 • **P:** 960 + 150 = 1110	**Iteration 14: Node P** • **M:** 1110 + 150 = 1260 • **O:** 1110 + 200 = 1310 • **R:** 1110 + 200 = 1310
Iteration 15: Node N • **K:** 1220 + 350 = 1570 • **O:** 1220 + 300 = 1520	**Iteration 16: Node O** • **N:** 1310 + 300 = 1610 • **P:** 1310 + 200 = 1510 • **Q:** 1310 + 110 = 1420

Iteration 17: Node R • **P:** 1310 + 200 = 1510 • **T:** 1310 + 160 = 1470	**Iteration 18: Node Q** • **O:** 1420 + 110 = 1530 • **S:** 1420 + 80 = 1500
Iteration 19: Node T • **R:** 1470 + 160 = 1630 • **S:** 1470 + 60 = 1530	**Iteration 20: Node S** • **Q:** 1500 + 80 = 1580 • **T:** 1500 + 60 = 1560

From source **A** to sink **T** the shortest path is

A → E → G → J → L → M → P → R → T = 1470KM

QUESTION FOUR

Refer to the code in github.

QUESTION FIVE

Iteration 1: Select Node L (dist 0) • **J:** 0 + 90 = 90 • **M:** 0 + 100 = 100 • **K:** 0 + 120 = 120 • **I:** 0 + 200 = 200	**Iteration 2: Select Node J (dist 90)** • **H:** 90 + 50 = 140 • **K:** 90 + 100 = 190 (keep 120) • **G:** 90 + 120 = 210 • **I:** 90 + 300 = 390 (keep 200)
Iteration 3: Select Node M (dist 100) • **K:** 100 + 110 = 210 (keep 120) • **P:** 100 + 150 = 250	**Iteration 4: Select Node K (dist 120)** • **F:** 120 + 150 = 270 • **N:** 120 + 350 = 470
Iteration 5: Select Node H (dist 140) • **G:** 140 + 80 = 220 (keep 210) • **F:** 140 + 160 = 300 (keep 270)	**Iteration 6: Select Node I (dist 200)** • **G:** 200 + 200 = 400 (keep 210) • **M:** 200 + 100 = 300 (keep 100)
Iteration 7: Select Node G (dist 210) • **F:** 210 + 100 = 310 (keep 270) • **E:** 210 + 200 = 410	**Iteration 8: Select Node P (dist 250)** • **R:** 250 + 200 = 450 • **O:** 250 + 200 = 450
Iteration 9: Select Node F (dist 270) • **D:** 270 + 350 = 620	**Iteration 10: Select Node E (dist 410)** • **C:** 410 + 200 = 610 • **D:** 410 + 80 = 490 • **A:** 410 + 450 = 860
Iteration 11: Select Node R (dist 450) • **T:** 450 + 160 = 610	**Iteration 12: Select Node O (dist 450)** • **Q:** 450 + 110 = 560 • **N:** 450 + 300 = 750 (keep 470)
Iteration 13: Select Node N (dist 470) • **O:** 470 + 300 = 770 (keep 450)	**Iteration 14: Select Node D (dist 490)** • **C:** 490 + 150 = 640 (keep 610)
Iteration 15: Select Node Q (dist 560) • **S:** 560 + 80 = 640	**Iteration 16: Select Node C (dist 610)** • **B:** 610 + 200 = 810 • **A:** 610 + 300 = 910 (keep 860)

Iteration 17: Select Node T (dist 610) [TARGET REACHED] • **S:** 610 + 60 = 670 (keep 640)	**Iteration 18: Select Node S (dist 640)** • **T:** 640 + 60 = 700 (keep 610)
Iteration 19: Select Node B (dist 810) • **A:** 810 + 100 = 910 (keep 860)	**Iteration 20: Select Node A (dist 860)** • **—:** All nodes processed

L → M → P → R → T = 610

Node	DIST	PRED
L (Lilongwe)	0	—
J (Dowa)	90	L
M (Dedza)	100	L
K (Salima)	120	L
H (Ntcheu)	140	J
I (Mchinji)	200	L
G (Kasungu)	210	J
P (Neno)	250	M
F (Nkhotakota)	270	K
E (Mzimba)	410	G
R (Chiradzulu)	450	P
O (Machinga)	450	P
N (Mangochi)	470	K
D (Nkhata Bay)	490	E
Q (Zomba)	560	O
C (Rumphi)	610	E
T (Blantyre)	610	R
S (Thyolo)	640	Q
B (Karonga)	810	C
A (Chitipa)	860	E

QUESTION SIX

Yes. A minimum spanning tree exists for any graph that is **connected, weighted and undirected**. Our graph meets all three conditions (every district can be reached from every other district), so it has an MST. If the graph were disconnected, we would only get a minimum spanning *forest*.

Properties: the MST has V − 1 = **19 edges**, no cycles, and connects all 20 districts with the smallest possible total weight.

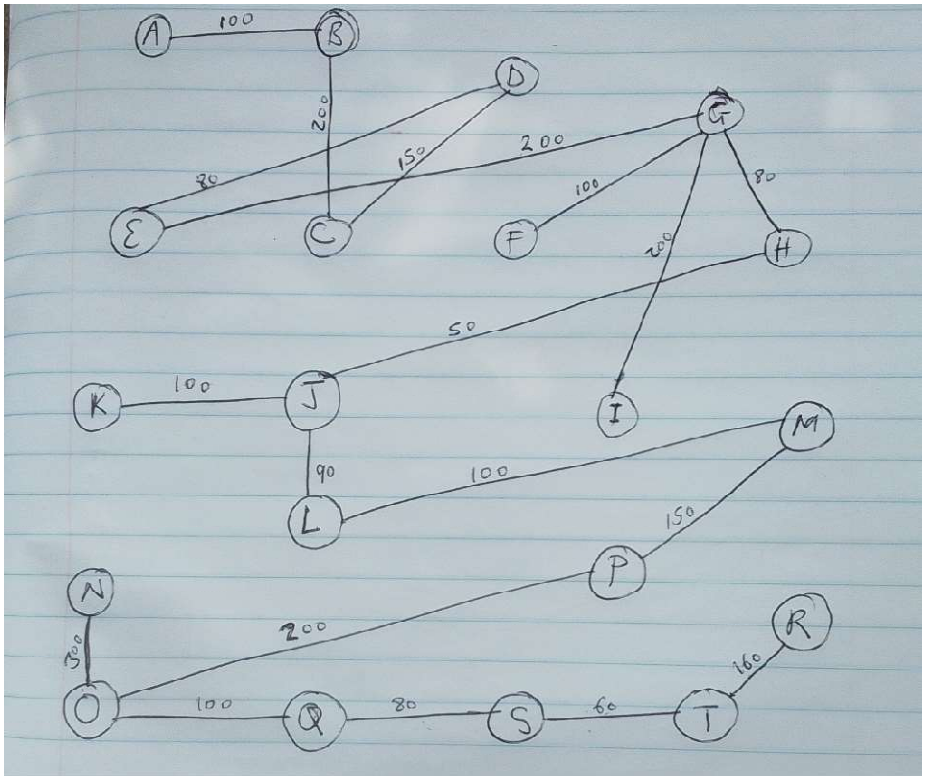
What it represents: the cheapest set of roads that keeps every district connected to every other district. In practice this is the road network that costs least to build or maintain (tarmac, patching, bridges), or the cheapest layout for a power line or fibre cable that must reach every district. It

tells us which 19 roads are essential and which 13 could be dropped without any district becoming cut off.

MST of our graph (found with Kruskal's algorithm):

Road	Between	Km
H – J	Ntchisi – Dowa	50
S – T	Chiradzulu – Blantyre	60
D – E	Nkhata Bay – Mzimba	80
G – H	Kasungu – Ntchisi	80
Q – S	Zomba – Chiradzulu	80
J – L	Dowa – Lilongwe	90
A – B	Chitipa – Karonga	100
F – G	Nkhotakota – Kasungu	100
J – K	Dowa – Salima	100
L – M	Lilongwe – Dedza	100
O – Q	Machinga – Zomba	110
C – D	Rumphi – Nkhata Bay	150
M – P	Dedza – Ntcheu	150
R – T	Neno – Blantyre	160
B – C	Karonga – Rumphi	200
E – G	Mzimba – Kasungu	200
G – I	Kasungu – Mchinji	200
O – P	Machinga – Ntcheu	200
N – O	Mangochi – Machinga	300

Total MST weight = 2510 km (19 edges).



Minimum spanning tree

Roads rejected because they would form a cycle (13): K – M (110), G – J (120), K – L (120), F – K (150), F – H (160), C – E (200), I – L (200), P – R (200), A – C (300), I – J (300), D – F (350), K – N (350) and A – E (450).

Note that an MST is **not** the same as the shortest-path tree from Dijkstra. The MST minimises total road length overall, while Dijkstra minimises the distance from one chosen source to each district. For example, the MST drops the roads A – E (450 km) and A – C (300 km), which Dijkstra uses for the shortest routes from Chitipa to Mzimba and to Rumphi.

7. Which MST Algorithm? (10 marks)

Kruskal's algorithm (with a Union-Find / disjoint-set structure).

sort all edges by weight, then go through them from smallest to largest, adding an edge if it connects two different components (checked with Union-Find) and skipping it otherwise, until V − 1 edges are chosen.

1. **The graph is sparse.** With E = 32 and V = 20 (only 32 of 190 possible roads), Kruskal runs in O(E log E), which is very small here. Prim's algorithm is better for dense graphs, where E is close to V^2.
2. **The data is already an edge list.** Roads are naturally stored as pairs of districts with a distance, as in the table in Question 1, so sorting them is easy.
3. **It is simple to implement and to trace by hand**, which helps with documentation and testing.
4. **It handles disconnected networks gracefully**, producing a forest when some districts are not linked.

Kruskal applied to our graph (edges sorted by weight; ties are taken in the order listed in Question 1):

Step	Road	km	Decision	Roads chosen
1	H – J (Ntchisi – Dowa)	50	Accept	1
2	S – T (Chiradzulu – Blantyre)	60	Accept	2
3	D – E (Nkhata Bay – Mzimba)	80	Accept	3
4	G – H (Kasungu – Ntchisi)	80	Accept	4
5	Q – S (Zomba – Chiradzulu)	80	Accept	5
6	J – L (Dowa – Lilongwe)	90	Accept	6
7	A – B (Chitipa – Karonga)	100	Accept	7
8	F – G (Nkhotakota – Kasungu)	100	Accept	8
9	J – K (Dowa – Salima)	100	Accept	9
10	L – M (Lilongwe – Dedza)	100	Accept	10
11	K – M (Salima – Dedza)	110	Reject (cycle)	10
12	O – Q (Machinga – Zomba)	110	Accept	11
13	G – J (Kasungu – Dowa)	120	Reject (cycle)	11
14	K – L (Salima – Lilongwe)	120	Reject (cycle)	11
15	C – D (Rumphi – Nkhata Bay)	150	Accept	12
16	F – K (Nkhotakota – Salima)	150	Reject (cycle)	12
17	M – P (Dedza – Ntcheu)	150	Accept	13
18	F – H (Nkhotakota – Ntchisi)	160	Reject (cycle)	13
19	R – T (Neno – Blantyre)	160	Accept	14
20	B – C (Karonga – Rumphi)	200	Accept	15
21	C – E (Rumphi – Mzimba)	200	Reject (cycle)	15
22	E – G (Mzimba – Kasungu)	200	Accept	16
23	G – I (Kasungu – Mchinji)	200	Accept	17
24	I – L (Mchinji – Lilongwe)	200	Reject (cycle)	17
25	O – P (Machinga – Ntcheu)	200	Accept	18

Step	Road	km	Decision	Roads chosen
26	P – R (Ntcheu – Neno)	200	Reject (cycle)	18
27	A – C (Chitipa – Rumphi)	300	Reject (cycle)	18
28	I – J (Mchinji – Dowa)	300	Reject (cycle)	18
29	N – O (Mangochi – Machinga)	300	Accept	19
30	D – F (Nkhata Bay – Nkhotakota)	350	Reject (cycle)	19
31	K – N (Salima – Mangochi)	350	Reject (cycle)	19
32	A – E (Chitipa – Mzimba)	450	Reject (cycle)	19

Thirteen roads are rejected because their two districts were already connected. The algorithm ends with 19 edges chosen (V − 1) and a total weight of 2510 km.

Comparison:

	Kruskal	Prim
Approach	Adds the globally cheapest safe edge	Grows one tree from a start vertex
Structure	Sort + Union-Find	Priority queue + adjacency list
Time	O(E log E)	O(E log V) with a heap
Best for	Sparse graphs	Dense graphs

Both would give the same total weight (2510 km). Equal-weight edges can be taken in any order, so ties could in principle give a different tree of the same total weight.

QUESTION 8

Deployment on github.